Category	Field Name	Data Type	Allowed Values	Condition / Context	Description	Source
Basics	enable	boolean	true, false	Inbound / Client configuration	Whether the inbound proxy or client entry is active and enabled.	1
Basics	port	integer	1-65535	Inbound / Outbound VLESS configuration	Network port on which the inbound proxy server listens or target server port number.	1, 2
Basics	protocol	string (enum)	vmess, vless, trojan, shadowsocks, wireguard, hysteria, http, mixed, tunnel, tun, mtproto, amneziawg, tuic, masque	Inbound configuration	The inbound proxy protocol type.	1
Basics	remark	string	Any valid string	Inbound / Host configuration	Human-readable label or display tag.	1
Basics	expiryTime	integer (int64)	Unix millisecond timestamp, 0 (unlimited), negative (wait for activation)	Client / Inbound configuration	Expiration timestamp after which access is disabled.	1
Basics	totalGB	integer (int64)	Non-negative integer, 0 (unlimited)	Client configuration	Total data traffic quota allocated to a client in Gigabytes.	1
Basics	email	string	Unique string email identifier	Client identification	Unique identifier for client accounts across attached inbounds.	1
Basics	subId	string	Unique alphanumeric string	Client subscription	Unique client subscription identifier used to pull client configurations.	1
Basics	limitIp	integer	Non-negative integer, 0 (unlimited)	Client IP limiting (requires Fail2ban)	Maximum number of distinct concurrent IP addresses allowed per client.	1
Basics	limitHwid	integer	Non-negative integer, 0 (unlimited)	Client device control	Maximum number of distinct hardware devices allowed to connect.	1
Basics	DNS Lua Script	string	Valid Lua script code	Xray-core v26.10.10 DNS configuration	Allows dynamic, programmable DNS resolution logic based on real-time metrics.	3
Basics	Routing Lua Script	string	Valid Lua script code	When populated, all legacy static UI routing rules are completely bypassed	Provides dynamic, programmatic evaluation for complex traffic routing topologies.	3
Basics	XUI_NONINTERACTIVE	integer / boolean	1	Unattended installation/provisioning scripts	Executes non-interactive installation with zero prompts, auto-generating randomized admin credentials and paths.	3
Protocol Settings	flow	string	xtls-rprx-vision, xtls-rprx-vision-udp443, none, ""	VLESS protocol / XTLS-Vision / Inbound & Outbound VLESS Client	Flow control scheme applied for XTLS-Vision acceleration and masking. Must match server inbound flow if specified in outbound.	1, 2
Protocol Settings	auth	string	Any valid string	Hysteria protocol	Authentication password or secret string used for Hysteria inbound.	1
Protocol Settings	password	string	Valid string key / password	Trojan / Shadowsocks / TUIC protocols	Client authentication password or pre-shared PSK.	1
Protocol Settings	uuid / id	string	Valid UUID v4 string, UUID generated by './xray uuid', or 1-30 byte string	VLESS / VMess / TUIC protocols, Inbound/Outbound VLESS client	Unique client identifier secret or authentication ID.	1, 2
Protocol Settings	allowedIPs	array of strings	Valid IPv4/IPv6 CIDRs (e.g. 10.0.0.2/32)	WireGuard / AmneziaWG protocols	IP address range assigned to a specific WireGuard/AmneziaWG peer.	1
Protocol Settings	forwardedPorts	string	Port list / range string (e.g. "80,443,8000-8100")	AmneziaWG protocol	AmneziaWG per-client port-forwarding specification.	1
Protocol Settings	decryption	string	none	Inbound VLESS settings	Decryption setting for VLESS inbound.	2
Protocol Settings	address	string	Domain name or IP address of server	Outbound VLESS vnext configuration	Server address.	2
Protocol Settings	encryption	string	none	VLESS outbound user settings / JSON subscription generation	Encryption method for VLESS outbound user. Explicitly emits key-value pair 'encryption': 'none' instead of empty string for parsing compatibility.	2, 3
Protocol Settings	TUN Windows Adapter Type	string	Not in source	TUN inbound on Windows operating systems	Explicitly configures the Windows adapter type to resolve hypervisor networking compatibility issues.	3
Protocol Settings	Finalmask Noise Reset Range	string	Range of byte sizes (e.g., 60-120 bytes)	Finalmask obfuscation settings	Defines precise noise reset byte ranges for packet size obfuscation.	3
Protocol Settings	Shadowsocks-2022 Cipher	string	2022-blake3-aes-256-gcm (Deprecated: 2022-blake3-chacha20-poly1305)	Shadowsocks-2022 inbounds	Specifies the encryption cipher for Shadowsocks-2022; legacy ChaCha20-Poly1305 is automatically migrated.	3
Security Schemes (REALITY/TLS)	security	string (enum)	none, tls, reality, same, auto, aes-128-gcm	Inbound / Host transport security	Transport layer security scheme applied to connections.	1
Security Schemes (REALITY/TLS)	sni / serverName / serverNames	string / array of strings	Valid domain name, list of domain names (wildcard "" not supported)	TLS / REALITY configurations, Inbound & Outbound REALITY	Server Name Indication used to mimic legitimate target server certificates or specify available SNIs usable by clients.	1, 2
Security Schemes (REALITY/TLS)	alpn	array of strings	h2, http/1.1, h3	TLS / REALITY / TUIC configurations	Application-Layer Protocol Negotiation list offered during TLS handshakes.	1
Security Schemes (REALITY/TLS)	pinnedPeerCertSha256	array of strings	Hex-encoded SHA-256 certificate hashes	TLS client certificate pinning	Public key/certificate SHA-256 hash list enforced for peer validation.	1
Security Schemes (REALITY/TLS)	fingerprint	string	chrome, firefox, safari, ios, android, edge, 360, qq, randomized	uTLS fingerprinting / REALITY Outbound	uTLS Client Hello fingerprint simulation to prevent active probing. Defaults to 'chrome' in REALITY outbound.	1, 2
Security Schemes (REALITY/TLS)	echConfigList	string	Base64 encoded ECH configuration list string	Encrypted Client Hello (ECH)	Encrypted Client Hello configuration payload for obfuscating TLS SNI.	1
Security Schemes (REALITY/TLS)	Use system CA	boolean	true, false	Outbound TLS settings	Toggles reliance on host OS certificate trust store instead of internally bundled Mozilla root certificates.	3
Security Schemes (REALITY/TLS)	NODE_TOKEN_ENCRYPTION	string	required	Master-Node fleet synchronization	Encrypts node API tokens at rest using base64 32-byte active keys in a locked JSON keyring file (mode 0600).	3
Security Schemes (REALITY/TLS)	Finalmask Passwords (Salamander / Gecko)	string	Minimum 4 bytes length	Finalmask configuration save operation	Cryptographic passwords failing to meet minimum length entropy are actively rejected.	3
Security Schemes (REALITY/TLS)	xtls-rprx-vision Flow	string	xtls-rprx-vision	Exclusive to transports where Xray natively executes TLS handshake (e.g., TLS/REALITY on raw TCP, VLESS native)	Enforces Vision flow restrictions to prevent transport incompatibility errors.	3
Stream Transports / Security (REALITY Inbound/Outbound)	show	boolean	true, false	Optional (REALITY Inbound / Outbound)	If true, outputs debug information.	2
Stream Transports / Security (REALITY Inbound)	target	string	host:port (e.g., example.com:443)	Required; same format as VLESS fallbacks dest	Target destination for fallback traffic.	2
Stream Transports / Security (REALITY Inbound)	xver	integer	PROXY protocol version (e.g., 0)	Optional; same format as VLESS fallbacks xver	PROXY protocol version for fallback connection.	2
Stream Transports / Security (REALITY Inbound)	privateKey	string	x25519 private key string generated by './xray x25519'	Required	Server side x25519 private key.	2
Stream Transports / Security (REALITY Inbound)	minClientVer	string	Version string in format x.y.z	Optional	Minimum allowed client Xray version.	2
Stream Transports / Security (REALITY Inbound)	maxClientVer	string	Version string in format x.y.z	Optional	Maximum allowed client Xray version.	2
Stream Transports / Security (REALITY Inbound)	maxTimeDiff	integer	Time difference in milliseconds	Optional	Maximum allowed time difference between client and server.	2
Stream Transports / Security (REALITY Inbound/Outbound)	shortIds / shortId	string / array of strings	Hex strings from '0' to 'f' with even length up to 16, or empty string ''	Required in inbound; Outbound must specify one configured shortId	List of available shortids usable by clients to distinguish different clients.	2
Stream Transports / Security (REALITY Inbound)	mldsa65Seed	string	Seed string generated by './xray mldsa65'	Optional	Seed for post-quantum quantum-resistant additional certificate signing.	2
Stream Transports / Security (REALITY Inbound Fallback Limit)	limitFallbackUpload.afterBytes	integer	Non-negative integer	Optional; bytesPerSec defaults to 0 (disabled)	Starts rate limiting fallback upload after transmitting specified bytes.	2
Stream Transports / Security (REALITY Inbound Fallback Limit)	limitFallbackUpload.bytesPerSec	integer	Non-negative integer (bytes/sec)	Optional; defaults to 0 (disabled)	Baseline rate for fallback upload speed limiting.	2
Stream Transports / Security (REALITY Inbound Fallback Limit)	limitFallbackUpload.burstBytesPerSec	integer	Non-negative integer (bytes/sec)	Optional; takes effect when greater than bytesPerSec	Burst rate for fallback upload speed limiting.	2
Stream Transports / Security (REALITY Inbound Fallback Limit)	limitFallbackDownload.afterBytes	integer	Non-negative integer	Optional; bytesPerSec defaults to 0 (disabled)	Starts rate limiting fallback download after transmitting specified bytes.	2
Stream Transports / Security (REALITY Inbound Fallback Limit)	limitFallbackDownload.bytesPerSec	integer	Non-negative integer (bytes/sec)	Optional; defaults to 0 (disabled)	Baseline rate for fallback download speed limiting.	2
Stream Transports / Security (REALITY Inbound Fallback Limit)	limitFallbackDownload.burstBytesPerSec	integer	Non-negative integer (bytes/sec)	Optional; takes effect when greater than bytesPerSec	Burst rate for fallback download speed limiting.	2
Stream Transports / Security (REALITY Outbound)	password	string	Public key generated from server's privateKey	Outbound REALITY settings	Public key acting as password for client authentication.	2
Stream Transports / Security (REALITY Outbound)	mldsa65Verify	string	Public key generated from server's mldsa65Seed	Optional	Public key for post-quantum quantum-resistant additional certificate verification.	2
Stream Transports / Security (REALITY Outbound)	spiderX	string	URL path and query parameters string (e.g. '/...')	Optional; recommended to be different for each client	Initial crawler path and parameters.	2
Stream Transports	XDRIVE	string	Google Drive, Localized web server folder, Templated HTTP storage API	Asynchronous Cloud-Decoy Transport	Routes proxy streams asynchronously through intermediary file storage systems and HTTP storage APIs.	3
Stream Transports	MASQUE	string	CONNECT-IP over HTTP/3 (QUIC), CONNECT-IP over HTTP/2, warp-masque outbound	Requires dedicated email and password client credentials	Tunnels raw IP packets or UDP datagrams within secure HTTP/3 or HTTP/2 connections.	3
Stream Transports	udphop mask	string	Valid UDP port hopping mask definition	Hysteria and XHTTP (HTTP/3) outbounds in JSON payload	Persists UDP port-hopping client settings at the end of the JSON payload to evade port-volume firewall blocking.	3
Stream Transports	TLS ALPN (WebSocket)	string	http/1.1	WebSocket stream transport default	Defaults ALPN selection to http/1.1 to prevent WebSocket handshake failures over certain routing paths.	3
Advanced Socket Options	finalMask	string (JSON object)	Valid JSON string specifying tcp/udp/quicParams masks	Xray finalmask noise customization	Customized stream noise mask configuration applied to underlying sockets.	1
Advanced Socket Options	mihomoIpVersion	string (enum)	dual, ipv4, ipv6, ipv4-prefer, ipv6-prefer	Clash / Mihomo client subscriptions	Target IP version resolution preference for client configurations.	1
Advanced Socket Options	vlessRoute	string	Integer string 0-65535 or empty	VLESS route steering	Steering value embedded into subscription UUID bytes for server-side packet routing.	1
Advanced Socket Options	UMask	string	0077	3X-UI systemd service configuration	Restricts default file creation permissions so panel/Xray files are readable/writable exclusively by the service account.	3
Advanced Socket Options	ReadWritePaths	string	Valid filesystem directory paths	Defined within custom systemd drop-in file (20-local.conf)	Explicitly overrides read-only system tree restrictions for designated filesystem paths.	3
Supported Architectures	Architecture Support	string	amd64, 386, arm64 (aarch64), armv7, armv6, armv5, s390x	Core platform deployment architectures	Hardware processor architectures natively supported by 3X-UI.	3
Deployment Environments	Environment Type	string	Bare Metal, Virtual Machines, Docker Containers (docker compose), Cloud-init Unattended Provisioning, Terraform (terraform-provider-3x-ui)	System and infrastructure deployment	Supported operating environments and provisioning systems.	3
Database Backends	Database Provider	string	SQLite (Default embedded), PostgreSQL 18 (Local/Remote), High-Availability Clusters, Distributed Node Sync	Data persistence layer	Supported database engines for storing panel configurations and traffic telemetry.	3